

John Liu

j369liu@uwaterloo.ca | john-liu.dev | github.com/johnroo2 | linkedin.com/in/john--liu

EDUCATION

University of Waterloo

Waterloo, ON

Computer Science, 3.9 Major GPA

Expected April 2028

- Admansi Entrance Scholarship (\$5,000)
- President's Scholarship of Distinction

WORK EXPERIENCE

Figma

San Francisco, CA

Software Engineer Intern

May 2026 – Aug 2026

- Developed editor and rendering infrastructure in **C++**, **TypeScript**, and **Rails** for Figma's design systems editor, enabling efficient composition of design variables such as colors, opacity, and typography attributes

WindBorne Systems Inc.

Palo Alto, CA

Software Engineer Intern

May 2025 – Aug 2025

- Engineering DRI for a meteorological insights platform used by energy traders, building forecasting and visualization features across **TypeScript**, **Rails**, **Rust**, and **WebGL** systems contributing to **\$2.5M ARR**
- Implemented tropical cyclone forecasting algorithms in **C++** and **Python** over **200TB+** of monthly global weather data, reducing query latency by 28% through a custom geospatial caching layer
- Contributed to a **Rust/WebGL** rendering engine powering interactive weather visualizations, maintaining interactive frame rates while rendering millions of forecast data points in the browser

CIBC

Toronto, ON

Software Engineer Intern

Jan 2025 – Apr 2025

- Built real-time chatbot features using **Python**, **Redis**, and websocket streaming, enabling low-latency tool execution and interactive user experiences for CIBC's internal AI assistant
- Delivered LLM-powered capabilities including **RAG** over patent databases, live news ingestion pipelines, and sentiment classification systems for document triage and automated summarization
- Architected a full-stack LLM prototyping framework (**Next.js**, **Prisma**, **FastAPI**) adopted across **10+** internal repositories, standardizing deployment and evaluation workflows

PROJECTS

SlimeScholars | React, Next.js, MongoDB, AWS

- Built an educational content platform with a custom lesson builder supporting **LaTeX**, **XML**, and **custom markdown**-based authoring workflows, with analytics dashboards tracking student engagement across lessons

Marble Investments | TypeScript, React, Next.js, MongoDB

- Created a **Stripe**-powered investment platform supporting fund subscriptions, redemptions, and investor onboarding with elliptic-curve cryptography for sensitive user data; helped scale assets under management from **\$250K** to **\$1.8M**

Firepit | React, Express

- Developed a **React/Express** coworking platform enabling live screen-sharing and collaboration, with type-safe **OpenAPI**-generated clients for real-time websocket and **REST** communication

unfollowed.lol | JavaScript, PostgreSQL, Django

- Launched a Chrome extension for tracking marketing analytics that reached **#8** on Product Hunt with **100+** upvotes and peaked at **4,000+** daily active users

SKILLS

Languages: C++, Python, TypeScript, JavaScript, Go, Java, C#, Ruby, Scala, SQL

Frameworks: React, Next.js, Rails, FastAPI, Spring Boot, PostgreSQL, MySQL, Redis, MongoDB, AWS, Azure

Areas: Distributed Systems, Computer Graphics, WebGL, Algorithms & Data Structures, Networking, Concurrency, Operating Systems